Mathematical Model Report: Spiral Vector Field Fitting

Rishabh Kumar

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Abstract

This report details the statistical formulation for fitting a parametric spiral vector field to noisy angular data. We derive the vector field from a scalar potential function, analyze the heteroskedastic noise structure induced by the Cartesian-to-Polar transformation, and formulate a Weighted Least Squares (WLS) objective function to estimate the model parameters.

1 Introduction

The objective of this model is to recover the parameters of a spiral vector field given a set of spatial observations. Unlike standard regression problems where errors are often assumed to be homoskedastic and Euclidean, this problem involves directional data where the noise variance depends on the magnitude of the underlying field.

2 Data and Preprocessing

The dataset consists of spatial coordinates $Z_i = (x_i, y_i)$ and observed vectors $W_i \in \mathbb{R}^2$. Preprocessing involves converting Cartesian vector observations into angular measurements $\hat{\alpha}_i$ and calculating the radial distances $r_i = \|Z_i\|$ for each observation point.

3 Model Specification

3.1 Parametric Spiral Vector Field

We define the vector field F_θ as the gradient of a scalar potential function $V_\theta(r, \phi)$, defined in polar coordinates (r, ϕ). The potential is given by:

$$V_\theta(r, \phi) = A\, g_p(r) - b\, \phi, \tag{1}$$

where $A, b \in \mathbb{R}$ are coefficients, and the radial function $g_p(r)$ is defined as:

$$g_p(r) = \begin{cases} \ln r, & p = 0, \\ \dfrac{r^p}{p}, & p \neq 0. \end{cases} \tag{2}$$

The parameter set is $\theta = \{A, b, p\}$. The induced vector field is the gradient of the potential. In polar coordinates, the gradient operator is $\nabla = \mathbf{e}_r \frac{\partial}{\partial r} + \mathbf{e}_\phi \frac{1}{r} \frac{\partial}{\partial \phi}$. Thus:

$$\begin{aligned} F_\theta(r, \phi) &= \nabla V_\theta(r, \phi) \\ &= \left(\frac{\partial}{\partial r} \left(A g_p(r) - b\phi \right) \right) \mathbf{e}_r + \left(\frac{1}{r} \frac{\partial}{\partial \phi} \left(A g_p(r) - b\phi \right) \right) \mathbf{e}_\phi \\ &= A g_p'(r) \mathbf{e}_r - \frac{b}{r} \mathbf{e}_\phi. \end{aligned} \tag{3}$$

Note that for all p, $g_p'(r) = r^{p-1}$. The magnitude of the field, denoted $\rho(r)$, is:

$$\rho(r) = \|F_\theta(r, \phi)\| = \sqrt{A^2 \left(r^{p-1}\right)^2 + \frac{b^2}{r^2}}. \tag{4}$$

Spiral Type	$g_p'(r)$	Vector Field $F(r, \phi)$	Magnitude $\rho(r)$	Weight $w(r) = \rho(r)^2$	Weighted LS Objective $L(\theta)$
Log Spiral	$\frac{1}{r}$	$F_{\text{log}}(r, \phi) = \left(\frac{A}{r}, -\frac{b}{r}\right)$	$\rho_{\text{log}}(r) = \frac{\sqrt{A^2 + b^2}}{r}$	$w_{\text{log}}(r) = \frac{A^2 + b^2}{r^2}$	$L_{\text{log}}(\theta) = \sum_i \frac{A^2 + b^2}{r_i^2} d^2(\hat{\alpha}_i, \alpha_\theta)$
					$= \frac{1}{b^2} \sum_i \frac{1/k^2 + 1}{r_i^2} d^2(\hat{\alpha}_i, \alpha_\theta) = L_{\text{log}}(k)$
Archimedean Spiral	1	$F_{\text{arch}}(r, \phi) = \left(A, -\frac{b}{r}\right)$	$\rho_{\text{arch}}(r) = \sqrt{A^2 + \frac{b^2}{r^2}}$	$w_{\text{arch}}(r) = A^2 + \frac{b^2}{r^2}$	$L_{\text{arch}}(\theta) = \sum_i \left(A^2 + \frac{b^2}{r_i^2}\right) d^2(\hat{\alpha}_i, \alpha_\theta)$
Fermat Spiral	r	$F_{\text{fermat}}(r, \phi) = \left(Ar, -\frac{b}{r}\right)$	$\rho_{\text{fermat}}(r) = \sqrt{A^2 r^2 + \frac{b^2}{r^2}}$	$w_{\text{fermat}}(r) = A^2 r^2 + \frac{b^2}{r^2}$	$L_{\text{fermat}}(\theta) = \sum_i \left(A^2 r_i^2 + \frac{b^2}{r_i^2}\right) d^2(\hat{\alpha}_i, \alpha_\theta)$

Table 1: Comparison of Vector Fields and Objective Functions for Different Spiral Types

3.2 Observation Model and Noise Structure

We observe the vector field at discrete spatial points Z_i for $i = 1, \dots, n$. The observed vector W_i is the true field perturbed by isotropic Gaussian noise:

$$W_i = F_{\theta_0}(Z_i) + \varepsilon_i, \tag{5}$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_2), \tag{6}$$

where I_2 is the 2×2 identity matrix.

Our primary quantity of interest is the direction of the field. The observed angle $\hat{\alpha}_i$ is:

$$\hat{\alpha}_i = \operatorname{atan2}(W_{i,2},\ W_{i,1}). \tag{7}$$

By linearizing the transformation from Cartesian vectors to polar angles, we find that the angular noise is heteroskedastic. Specifically, the relationship between the observed angle and the true angle $\alpha_{\theta_0}(Z_i)$ is:

$$\hat{\alpha}_i \approx \alpha_{\theta_0}(Z_i) + \eta_i, \tag{8}$$

where the variance of the angular error η_i is inversely proportional to the squared magnitude of the field at that point:

$$\operatorname{Var}(\eta_i) \approx \frac{\sigma^2}{\rho(Z_i)^2}. \tag{9}$$

Theorem 1 (Density of the angle $\Theta = \operatorname{atan2}(W_2, W_1)$ for an isotropic Gaussian)**.** *Let* $W = (W_1, W_2)^\top \sim \mathcal{N}_2(\mu, \sigma^2 I_2)$ *with* $\sigma > 0$ *and* $\mu \in \mathbb{R}^2$*. Define*

$$\Theta := \operatorname{atan2}(W_2, W_1) \in (-\pi, \pi].$$

Write $m := \|\mu\|$ *and* $\phi_0 := \operatorname{atan2}(\mu_2, \mu_1)$*, so that* $\mu = m(\cos\phi_0, \sin\phi_0)^\top$*. Then* Θ *has a density on* $(-\pi, \pi]$ *given by*

$$f_\Theta(\theta) = \frac{1}{2\pi} \exp\left(-\frac{m^2}{2\sigma^2}\right) + \frac{m\cos(\theta - \phi_0)}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{m^2 \sin^2(\theta - \phi_0)}{2\sigma^2}\right) \Phi\left(\frac{m\cos(\theta - \phi_0)}{\sigma}\right), \tag{10}$$

$$\theta \in (-\pi, \pi], \tag{11}$$

where Φ *denotes the standard normal cumulative distribution function.*

Proof. Let $w = (w_1, w_2)^\top \in \mathbb{R}^2$. The density of W is

$$f_W(w) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\|w-\mu\|^2}{2\sigma^2}\right).$$

Introduce polar coordinates $w = (r\cos\theta, r\sin\theta)^\top$ with $r \geq 0$ and $\theta \in (-\pi, \pi]$, so that $\Theta = \theta$ and the Jacobian determinant is r. Hence the joint density of (R, Θ) is

$$f_{R,\Theta}(r,\theta) = f_W(r\cos\theta, r\sin\theta)\, r = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\|(r\cos\theta, r\sin\theta) - \mu\|^2}{2\sigma^2}\right) r.$$

Write $\mu = m(\cos\phi_0, \sin\phi_0)^\top$. Then

$$\begin{aligned} \|(r\cos\theta, r\sin\theta) - \mu\|^2 &= r^2 + m^2 - 2rm(\cos\theta\cos\phi_0 + \sin\theta\sin\phi_0) \\ &= r^2 + m^2 - 2rm\cos(\theta - \phi_0). \end{aligned}$$

Set $c(\theta) := \cos(\theta - \phi_0)$ and rewrite

$$r^2 + m^2 - 2rm\cos(\theta-\phi_0) = (r - m\cos(\theta-\phi_0))^2 + m^2(1 - c(\theta)^2) = (r - m\cos(\theta-\phi_0))^2 + m^2\sin^2(\theta-\phi_0).$$

Therefore,

$$f_\Theta(\theta) = \int_0^\infty f_{R,\Theta}(r,\theta)\, dr = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{m^2\sin^2(\theta-\phi_0)}{2\sigma^2}\right) \int_0^\infty r\exp\left(-\frac{(r - m\cos(\theta-\phi_0))^2}{2\sigma^2}\right) dr.$$

Let $a := m\cos(\theta - \phi_0)$ and substitute $u = (r-a)/\sigma$ (so $r = \sigma u + a$, $dr = \sigma du$), giving

$$\begin{aligned} \int_0^\infty r\exp\left(-\frac{(r-a)^2}{2\sigma^2}\right) dr &= \int_{-a/\sigma}^\infty (\sigma u + a) e^{-u^2/2}\, \sigma du \\ &= \sigma^2 \int_{-a/\sigma}^\infty u e^{-u^2/2}\, du \; + \; a\sigma \int_{-a/\sigma}^\infty e^{-u^2/2}\, du. \end{aligned}$$

The first integral evaluates to $e^{-a^2/(2\sigma^2)}$ since $\int u e^{-u^2/2} du = -e^{-u^2/2}$, hence

$$\int_{-a/\sigma}^\infty u e^{-u^2/2}\, du = \left[-e^{-u^2/2}\right]_{-a/\sigma}^\infty = e^{-a^2/(2\sigma^2)}.$$

For the second integral,

$$\int_{-a/\sigma}^\infty e^{-u^2/2}\, du = \sqrt{2\pi}\,\Phi\left(\frac{a}{\sigma}\right),$$

by the definition of Φ and symmetry of the standard normal density. Combining,

$$\int_0^\infty r\exp\left(-\frac{(r-a)^2}{2\sigma^2}\right) dr = \sigma^2 e^{-a^2/(2\sigma^2)} + a\sigma\sqrt{2\pi}\,\Phi\left(\frac{a}{\sigma}\right).$$

Substituting back into $f_\Theta(\theta)$ with $a = m\cos(\theta - \phi_0)$ yields,

$$\begin{aligned} f_\Theta(\theta) &= \frac{1}{2\pi\sigma^2} \exp\left(-\frac{m^2\sin^2(\theta-\phi_0)}{2\sigma^2}\right) \left[\sigma^2 e^{-\frac{m^2\cos(\theta-\phi_0)^2}{2\sigma^2}} + (m\cos(\theta-\phi_0))\sigma\sqrt{2\pi}\,\Phi\left(\frac{m\cos(\theta-\phi_0)}{\sigma}\right)\right] \\ &= \frac{1}{2\pi}\exp\left(-\frac{m^2}{2\sigma^2}\right) + \frac{m\cos(\theta-\phi_0)}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{m^2\sin^2(\theta-\phi_0)}{2\sigma^2}\right) \Phi\left(\frac{m\cos(\theta-\phi_0)}{\sigma}\right), \end{aligned}$$

and recalling $c(\theta) = \cos(\theta - \phi_0)$ gives the stated formula. □

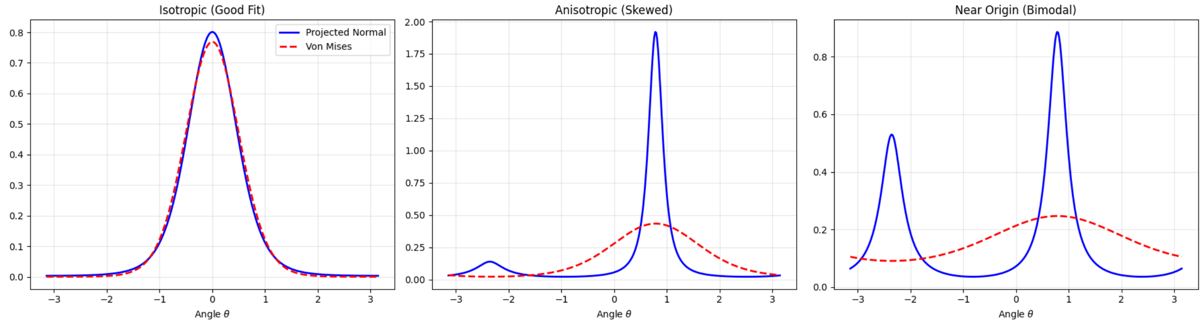

Figure 1: Von Mises as an approximation to the projected normal. The approximation is accurate in the isotropic regime but deteriorates under anisotropy and fails entirely when the projected normal becomes skewed or bimodal.

Maximizing the likelihood from Theorem 1 is complex due to the non-linear form of the density. However, for practical purposes, we focus on the variance structure derived from the linear approximation. Also, changing the distance structure (e.g., using sine-wrapped loss) leads to similar conclusions regarding the weighting of observations.

This implies that angular observations are less reliable in regions where the vector field magnitude $\rho(r)$ is small.

Most directional statistics regression assume the angular noise as homoskedastic (e.g., Von Mises distribution with constant concentration parameter). However, in our case, the noise variance depends on the field magnitude, necessitating a heteroskedastic treatment.

3.3 Traditional Approaches and Limitations

Standard regression techniques for directional data, such as those based on the Von Mises distribution, typically assume constant noise variance. Applying such methods directly to our data would ignore the critical dependence of angular uncertainty on the field magnitude, leading to biased parameter estimates.

Consider angular data, $\hat{\alpha}_1, \dots, \hat{\alpha}_n$ observed at locations $Z_1, \dots, Z_n$. The regression model is given by:

$$\hat{\alpha}_i = \alpha_\theta(Z_i) + \eta_i, \tag{12}$$

where η_i are independent errors. Traditional methods assume $\eta_i \sim \mathrm{VM}(0, \kappa)$ with constant concentration κ. We can write the joint likelihood as:

$$\ell(\theta|\hat{\alpha}_1, \dots, \hat{\alpha}_n) = \sum_{i=1}^{n} \left[\kappa \cos(\hat{\alpha}_i - \alpha_\theta(Z_i)) - \log I_0(\kappa)\right], \tag{13}$$

Theorem 2 (Angular distribution for a Gaussian-induced line field)**.** *Let $W = (W_1, W_2)^\top \sim \mathcal{N}_2(\mu, \sigma^2 I_2)$ with $\sigma > 0$ and $\mu \in \mathbb{R}^2$. Define the oriented angle*

$$\Theta := \mathrm{atan2}(W_2, W_1) \in (-\pi, \pi],$$

and the corresponding line-field angle

$$\Psi := \Theta \bmod \pi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Let $m := \|\mu\|$ and $\phi_0 := \mathrm{atan2}(\mu_2, \mu_1)$. Then Ψ admits a density on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right]$ given by

$$f_\Psi(\psi) = f_\Theta(\psi) + f_\Theta(\psi + \pi),$$

where f_Θ is the projected normal density

$$f_\Theta(\theta) = \frac{1}{2\pi}\exp\!\left(-\frac{m^2}{2\sigma^2}\right) + \frac{m\cos(\theta-\phi_0)}{\sqrt{2\pi}\,\sigma}\exp\!\left(-\frac{m^2\sin^2(\theta-\phi_0)}{2\sigma^2}\right)\Phi\!\left(\frac{m\cos(\theta-\phi_0)}{\sigma}\right).$$

The resulting distribution is π-periodic and symmetric, and hence appropriate for modeling line fields.

Theorem 3 (Density of the line-field angle $\Psi = \Theta \bmod \pi$)**.** *Let $W = (W_1, W_2)^\top \sim \mathcal{N}_2(\mu, \sigma^2 I_2)$ with $\sigma > 0$ and $\mu \in \mathbb{R}^2$. Define the oriented angle*

$$\Theta := \operatorname{atan2}(W_2, W_1) \in (-\pi, \pi],$$

and the line-field (axial) angle

$$\Psi := \Theta \bmod \pi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Let $m := \|\mu\|$ and $\phi_0 := \operatorname{atan2}(\mu_2, \mu_1)$. Then Ψ has density on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right]$ given by

$$f_\Psi(\psi) = f_\Theta(\psi) + f_\Theta(\psi+\pi), \qquad \psi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right],$$

where f_Θ is the projected normal density

$$f_\Theta(\theta) = \frac{1}{2\pi}\exp\!\left(-\frac{m^2}{2\sigma^2}\right) + \frac{m\cos(\theta-\phi_0)}{\sqrt{2\pi}\,\sigma}\exp\!\left(-\frac{m^2\sin^2(\theta-\phi_0)}{2\sigma^2}\right)\Phi\!\left(\frac{m\cos(\theta-\phi_0)}{\sigma}\right).$$

Moreover, f_Ψ admits the simplified closed form

$$\boxed{\begin{aligned} f_\Psi(\psi) &= \frac{1}{\pi}\exp\!\left(-\frac{m^2}{2\sigma^2}\right) \\ &\quad + \frac{m\cos(\psi-\phi_0)}{\sqrt{2\pi}\,\sigma}\exp\!\left(-\frac{m^2\sin^2(\psi-\phi_0)}{2\sigma^2}\right)\left[2\Phi\!\left(\frac{m\cos(\psi-\phi_0)}{\sigma}\right)-1\right], \end{aligned}} \qquad \psi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Proof. **Step 1 (folding identity).** Let $I := (-\pi, \pi]$ and $J := \left(-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Define the map $q : I \to J$ by $q(\theta) = \theta \bmod \pi$, i.e. $q(\theta) = \theta$ if $\theta \in J$ and $q(\theta) = \theta - \pi$ if $\theta \in (\frac{\pi}{2}, \pi]$ (and equivalently $q(\theta) = \theta + \pi$ if $\theta \in (-\pi, -\frac{\pi}{2}]$). Then q is measurable and each $\psi \in J$ has exactly two preimages in I, namely ψ and $\psi + \pi$.

For any Borel set $B \subseteq J$,

$$\mathbb{P}(\Psi \in B) = \mathbb{P}(q(\Theta) \in B) = \mathbb{P}(\Theta \in q^{-1}(B)).$$

Since $q^{-1}(B) = B \,\dot\cup\, (B + \pi)$ is a disjoint union (modulo boundary points of measure zero), we have

$$\mathbb{P}(\Psi \in B) = \mathbb{P}(\Theta \in B) + \mathbb{P}(\Theta \in B + \pi) = \int_B f_\Theta(\theta)\,d\theta + \int_{B+\pi} f_\Theta(\theta)\,d\theta.$$

In the second integral, substitute $\theta = \psi + \pi$ to obtain

$$\mathbb{P}(\Psi \in B) = \int_B \big(f_\Theta(\psi) + f_\Theta(\psi+\pi)\big)\,d\psi.$$

Thus Ψ has density

$$f_\Psi(\psi) = f_\Theta(\psi) + f_\Theta(\psi+\pi), \qquad \psi \in J.$$

Step 2 (simplification of the folded form). Let $u := \psi - \phi_0$ and abbreviate

$$c := \cos u, \qquad s := \sin u, \qquad x := \frac{mc}{\sigma}.$$

Using trigonometric identities,

$$\cos(\psi+\pi-\phi_0) = \cos(u+\pi) = -\cos u = -c, \qquad \sin^2(\psi+\pi-\phi_0) = \sin^2(u+\pi) = \sin^2 u = s^2.$$

Apply these to $f_\Theta(\psi+\pi)$:

$$f_\Theta(\psi+\pi) = \frac{1}{2\pi} e^{-m^2/(2\sigma^2)} + \frac{m(-c)}{\sqrt{2\pi}\sigma}\, e^{-m^2 s^2/(2\sigma^2)}\, \Phi\!\left(\frac{m(-c)}{\sigma}\right).$$

Therefore,

$$\begin{aligned} f_\Psi(\psi) &= \left(\frac{1}{2\pi} e^{-m^2/(2\sigma^2)} + \frac{mc}{\sqrt{2\pi}\sigma} e^{-m^2 s^2/(2\sigma^2)} \Phi(x)\right) + \left(\frac{1}{2\pi} e^{-m^2/(2\sigma^2)} - \frac{mc}{\sqrt{2\pi}\sigma} e^{-m^2 s^2/(2\sigma^2)} \Phi(-x)\right) \\ &= \frac{1}{\pi} e^{-m^2/(2\sigma^2)} + \frac{mc}{\sqrt{2\pi}\sigma} e^{-m^2 s^2/(2\sigma^2)} \big(\Phi(x) - \Phi(-x)\big). \end{aligned}$$

Finally, using the standard identity $\Phi(-x) = 1 - \Phi(x)$,

$$\Phi(x) - \Phi(-x) = \Phi(x) - (1 - \Phi(x)) = 2\Phi(x) - 1.$$

Substituting back $c = \cos(\psi - \phi_0)$ and $s = \sin(\psi - \phi_0)$ yields the claimed closed form. □

4 Estimation and Inference

4.1 Weighted Least Squares Objective

To estimate θ, we minimize the discrepancy between the observed angles and the model-predicted angles. A standard least squares approach is insufficient due to the heteroskedasticity described above. We instead employ a Weighted Least Squares (WLS) approach.

The negative log-likelihood (ignoring constant terms) suggests minimizing the variance-weighted squared error. Since angular data lies on a circle (manifold S^1), we use a wrapped loss function based on the sine of the angular difference.

The objective function $L(\theta)$ is:

$$L(\theta) = \sum_{i=1}^{n} \rho(Z_i)^2 \sin^2\big(\hat{\alpha}_i - \alpha_\theta(Z_i)\big). \tag{14}$$

Here, the weight $\rho(Z_i)^2$ naturally down-weights observations where the field magnitude is small (and thus angular uncertainty is high).

4.2 Optimization

The estimator $\hat{\theta}$ is obtained by:

$$\hat{\theta} = \arg\min_\theta L(\theta). \tag{15}$$

Since $L(\theta)$ is non-linear with respect to parameters p and the geometry of the angle α_θ, we utilize numerical optimization (e.g., L-BFGS-B or Newton-Raphson) to solve for $\hat{\theta}$.

4.3 Objective correction

As seen in Equation (14), the weights $\rho(Z_i)^2$ provide a natural correction for the heteroskedasticity in angular observations. But this is not enough. To see that, we have to look at the log-likelihood function more closely. Depending on the distance function used to measure the angular error, we can derive different forms of the log-likelihood function. Consider we observe the angles $\alpha_1, \ldots, \alpha_n$ at locations $Z_1, \ldots, Z_n$. Then the log-likelihood function can be derived as follows:

$$\ell(\theta \,|\, \alpha_1, \ldots, \alpha_n) = -\frac{1}{2} \sum_{i=1}^{n} \left[\log\!\left(2\pi \frac{\sigma^2}{\rho_\theta(Z_i)^2}\right) + \frac{d(\hat{\alpha}_i, \alpha_\theta(Z_i))^2 \rho_\theta(Z_i)^2}{\sigma^2}\right] \tag{16}$$

5 Results

[Insert parameter estimates for A, b, p here]

6 Discussion

The weighted loss function effectively handles the noise structure inherent in angular measurements of vector fields. [Discuss fit quality here].

7 Conclusion

We have successfully formulated a statistical model for spiral vector fields that accounts for the geometric properties of angular noise.

References